

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 1

Year: 2015

Subject: 392151 Chemistry I

Section: 15 - 16

Date: 14 October 2015

Time: 8.00 - 10.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. No calculators are allowed in the examination.
4. This exam is a closed book examination.
5. The examination has 7 pages (including this page), 2 sections and a total score of 60 points.

Score	Part 1	Part 2	Total
-------	--------	--------	-------



WebElements: the periodic table on the world-wide web

<http://www.webelements.com/>

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
hydrogen 1 H 1.0079																	helium 2 He 4.0026	
lithium 3 Li 6.941	beryllium 4 Be 9.0122											boron 5 B 10.811	carbon 6 C 12.011	nitrogen 7 N 14.007	oxygen 8 O 15.999	fluorine 9 F 18.998	neon 10 Ne 20.180	
sodium 11 Na 22.990	magnesium 12 Mg 24.305											aluminium 13 Al 26.982	silicon 14 Si 28.086	phosphorus 15 P 30.974	sulfur 16 S 32.065	chlorine 17 Cl 35.453	argon 18 Ar 39.948	
potassium 19 K 39.098	calcium 20 Ca 40.078		scandium 21 Sc 44.956	titanium 22 Ti 47.867	vanadium 23 V 50.942	chromium 24 Cr 51.996	manganese 25 Mn 54.938	iron 26 Fe 55.845	cobalt 27 Co 58.933	nickel 28 Ni 58.693	copper 29 Cu 63.546	zinc 30 Zn 65.39	gallium 31 Ga 69.723	germanium 32 Ge 72.61	arsenic 33 As 74.922	selenium 34 Se 78.96	bromine 35 Br 79.904	krypton 36 Kr 83.80
rubidium 37 Rb 85.468	strontium 38 Sr 87.62		yttrium 39 Y 88.906	zirconium 40 Zr 91.224	niobium 41 Nb 92.906	molybdenum 42 Mo 95.94	technetium 43 Tc [98]	ruthenium 44 Ru 101.07	rhodium 45 Rh 102.91	palladium 46 Pd 106.42	silver 47 Ag 107.87	cadmium 48 Cd 112.41	indium 49 In 114.82	tin 50 Sn 118.71	antimony 51 Sb 121.76	tellurium 52 Te 127.60	iodine 53 I 126.90	xenon 54 Xe 131.29
caesium 55 Cs 132.91	barium 56 Ba 137.33	57-70 *	lutetium 71 Lu 174.97	hafnium 72 Hf 178.49	tantalum 73 Ta 180.95	tungsten 74 W 183.84	rhenium 75 Re 186.21	osmium 76 Os 190.23	iridium 77 Ir 192.22	platinum 78 Pt 195.08	gold 79 Au 196.97	mercury 80 Hg 200.59	thallium 81 Tl 204.38	lead 82 Pb 207.2	bismuth 83 Bi 208.98	polonium 84 Po [209]	astatine 85 At [210]	radon 86 Rn [222]
francium 87 Fr [223]	radium 88 Ra [226]	89-102 **	lawrencium 103 Lr [262]	rutherfordium 104 Rf [261]	dubnium 105 Db [262]	seaborgium 106 Sg [266]	bohrium 107 Bh [264]	hassium 108 Hs [269]	meitnerium 109 Mt [268]	unnilium 110 Uun [271]	ununium 111 Uuu [272]	unubium 112 Uub [277]		ununquadium 114 Uuq [289]				

Key:

element name
atomic number
symbol
atomic weight (mean relative mass)

lanthanum 57 La 138.91	cerium 58 Ce 140.12	praseodymium 59 Pr 140.91	neodymium 60 Nd 144.24	promethium 61 Pm [145]	samarium 62 Sm 150.36	europium 63 Eu 151.96	gadolinium 64 Gd 157.25	terbium 65 Tb 158.93	dysprosium 66 Dy 162.50	holmium 67 Ho 164.93	erbium 68 Er 167.26	thulium 69 Tm 168.93	ytterbium 70 Yb 173.04
actinium 89 Ac [227]	thorium 90 Th 232.04	protactinium 91 Pa 231.04	uranium 92 U 238.03	neptunium 93 Np [237]	plutonium 94 Pu [244]	americium 95 Am [243]	curium 96 Cm [247]	berkelium 97 Bk [247]	californium 98 Cf [251]	einsteinium 99 Es [252]	fermium 100 Fm [257]	mendelevium 101 Md [258]	nobelium 102 No [259]

*lanthanoids

**actinoids

Symbols and names: the symbols and names of the elements, and their spellings are those recommended by the International Union of Pure and Applied Chemistry (IUPAC - <http://www.iupac.org/>). Names have yet to be proposed for the most recently discovered elements 110–112 and 114 so those used here are IUPAC's temporary systematic names. In the USA and some other countries, the spellings aluminium and caesium are normal while in the UK and elsewhere the common spelling is sulphur.

Group labels: the numeric system (1–18) used here is the current IUPAC convention.

Atomic weights (mean relative masses): Apart from the heaviest elements, these are the IUPAC 2001 values and given to 5 significant figures. Elements for which the atomic weight is given within square brackets have no stable nuclides and are represented by the element's longest lived isotope.

©2002 Dr Mark J Winter [WebElements Ltd and University of Sheffield, webelements@sheffield.ac.uk]. All rights reserved. For updates to this table see <http://www.webelements.com/webelements/support/media/pdf/>. Version date: 3 Apr 2002.

Part 1. Multiple Choice (20 points)

Choose the letter of the choice in the answer table that best completes the statement or answers the question.

Question no.	a.	b.	c.	d.
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

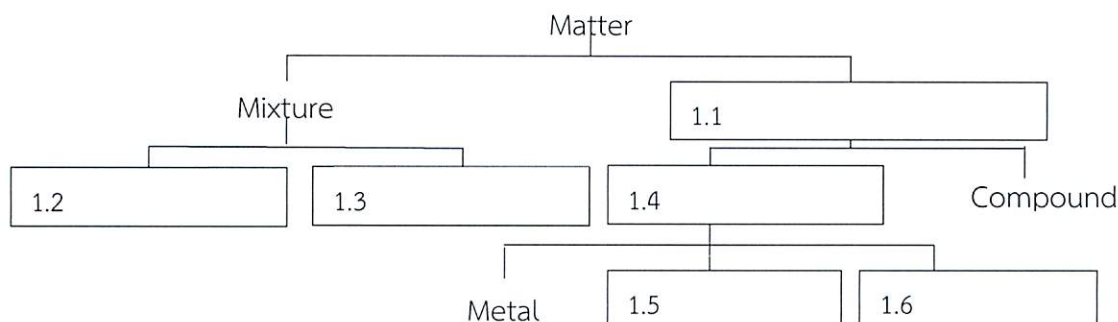
Question no.	a.	b.	c.	d.
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				

- In the mixture, the substance that make up the mixture:
 - are in a fixed ratio
 - keep their physical properties
 - cannot be separated by mechanical means
 - are only an element
- Metals like to _____ electrons to become _____ ions.
 - lose, positive
 - lose, negative
 - gain, positive
 - gain, negative
- Which two words apply to the sodium chloride (salt)?
 - liquid, mixture
 - solid, mixture
 - liquid, compound
 - solid, compound
- Which of the following can be classified as a mixture?
 - sugar
 - table salt
 - air
 - Iron
- Which of the following is a chemical change?
 - Ice starts to melt
 - A car starts to rust
 - melting butter for popcorn
 - Separating sand and gravel
- Which choice is not an example of pure substance?
 - milk shake
 - water
 - olive oil
 - Coca cola
- In any chemical reaction, the total mass of the reactants is always equal to the total mass of materials produced. This is known as the law of:
 - constant composition
 - conservation of energy
 - chemical reaction
 - conservation of mass
- What is an element?
 - a pure substance made up of one type of atom
 - a pure substance made up of 2 or more types of atoms
 - a homogeneous liquid
 - a heterogeneous solid

9. A decomposition reaction can be compared to _____.
- two dancing couples switching partners
 - dancing couple breaking up
 - two single people joining for a dance
 - None of these is correct
10. The mass of the atom is determined by _____.
- atomic mass number
 - neutron
 - electron
 - atomic number
11. Rutherford's atomic model was the first one to include _____.
- proton
 - nucleus
 - energy level
 - neutron
12. The cathode ray tube was instrumental in the discovery of:
- the electron
 - the neutron
 - the proton
 - the atom, which is mostly empty space
13. What is the key difference between the models of the atom proposed by Bohr and Electron cloud?
- Electron cloud model includes a nucleus, but Bohr's does not.
 - Electron cloud model has the atom made up of mostly empty space, but Bohr's does not.
 - Electron cloud model includes neutrons, but Bohr's does not.
 - Electron cloud model has orbitals, but Bohr's does not.
14. Which of the following atoms has 7 neutrons?
- $^{15}_{7}\text{N}$
 - $^{13}_{6}\text{C}$
 - $^{7}_{3}\text{Li}$
 - $^{5}_{2}\text{He}$
15. Which of the following are isotopes of $^{40}_{20}\text{Ca}$?
- $^{40}_{19}\text{K}$
 - $^{42}_{20}\text{Ca}$
 - $^{39}_{19}\text{K}$
 - $^{40}_{18}\text{Ar}$
16. The electron configuration for helium is $1s^2$. The "2" stands for _____.
- the orbital
 - how many electrons helium has
 - the energy level the electrons are in
 - the atomic number
17. According to problem 16, the "s" stands for _____.
- the orbital
 - how many electrons helium has
 - the energy level the electrons are in
 - the atomic number
18. The vertical columns on the periodic table are called _____.
- metals
 - non-metals
 - periods
 - group/families
19. Which element has the electron configuration $1s^2 2s^2 2p^5$?
- boron
 - nitrogen
 - fluorine
 - phosphorus
20. Which of the following material is a good thermal conductors?
- silver
 - concrete
 - water
 - snow

Part 2 Write your answer in the space provided for each question (40 points)

1. Complete the diagram for the classification of matter (3 points)



2. Classify each of the following as an element, a compound or a mixture. If it is a mixture, also say whether it is a homogeneous or heterogeneous mixture. (4 points)

- 2.1 pizza _____ 2.2 CO₂ _____
 2.3 carbon in pencil _____ 2.4 tap water _____
 2.5 air _____ 2.6 ice tea _____
 2.7 brass _____ 2.8 gold _____

3. Classify each of the following changes in matter as physical or chemical. (3 points)

- 3.1 sour milk _____ 3.2 warming food in microwave _____
 3.3 distillation _____ 3.4 dissolving instant coffee _____
 3.5 fireworks exploding _____ 3.6 burning paper _____

4. For each of the following reactions, say whether it is a synthesis or decomposition reaction and also say whether it is an endothermic or an exothermic reaction: (2 points).

- | | synthesis/ decomposition | endothermic/exothermic |
|--|--------------------------|------------------------|
| 4.1 $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ | _____ | _____ |
| 4.2 $\text{CH}_4 \rightarrow \text{C} + 2\text{H}_2$ | _____ | _____ |

5. Draw the following atomic model. (3 points).

5.1 Thomson's	5.2 Rutherford's	5.3 Bohr's

6. Calculate the frequency and energy of a photon of light of wavelength 660 nm. (4 points)
 (speed of light; $c = 3.0 \times 10^8$ m/s and Plank's constant; $h = 6.6 \times 10^{-34}$ J.s)

7. Complete the following table (Hint: using detail in the periodic table) (4 points)

Nuclear Symbol	Atomic number (Z)	Atomic mass (A)	Number of protons	Number of electrons	Number of neutrons
${}^{64}_{29}\text{Cu}^{2+}$					
	21				
		31		18	16
Zn				28	

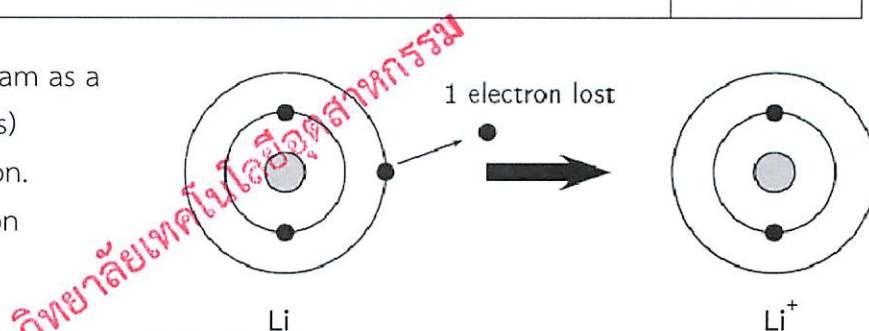
8. Write the detailed electron configurations and identify its location (period and group) in the periodic table for each of the element in table below. (5 points)

Element	Subshell	An atom of	Shell	Period number	Group number
${}_3\text{X}$ (3 electrons)	$1s^2 2s^1$	Li	2,1		
${}_{19}\text{X}$ (19 electrons)					
${}_{31}\text{X}$ (31 electrons)					

9. Write the detailed electron configurations and draw the electron diagram in orbitals of atoms and ion in the table (8 points)

Element	Subshell and electron diagram in orbitals	Shell
${}^3\text{Li}$ (Example)	$1s^2 \quad 2s^1$ 	2,1
${}^{11}\text{Na}$		
${}^{36}\text{Kr}$		
${}^{29}\text{Cu}$		
${}^{26}\text{Fe}^{2+}$		

10. Using the lithium diagram as a guide to draw (4 points)
 10.1 Fluorine and its ion.
 10.2 Calcium and its ion



Fluorine and its ion

Calcium and its ion